SSL/TLS Handshake and Certificate Analysis Report

Tool Used: Wireshark

1. Objective

To capture and analyze the SSL/TLS handshake process while accessing https://bluecloudsoftech.com using Wireshark, and to inspect the server SSL certificate details.

2. Tools Used

- Wireshark (Packet Analysis Tool)
- Web Browser (Chrome/Firefox)
- Windows Operating System

3. Procedure

1. Installed and opened Wireshark as administrator.
2. Selected active network interface (Wi-Fi).
3. Started packet capture.
4. Applied filter: tls.
5. Opened https://bluecloudsoftech.com in browser and logged in.
6. Captured SSL/TLS handshake packets including Client Hello, Server Hello, and Certificate exchange.
7. Stopped capture after successful login session.

4. Observations

During packet capture, the following TLS handshake stages were observed:

• TCP 3-way handshake (SYN, SYN-ACK, ACK)
• TLS Client Hello (supported cipher suites, TLS version, SNI)
• TLS Server Hello (selected cipher suite and TLS version)
• Certificate exchange (server certificate and intermediate CA)
• Secure encrypted communication established after handshake completion

5. SSL Certificate Analysis

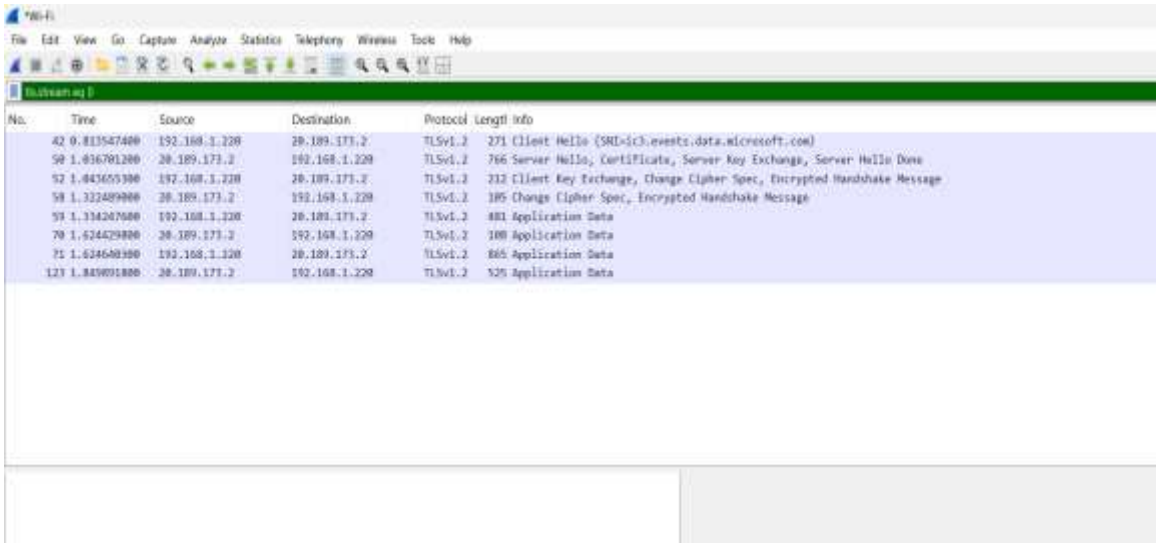
The server certificate was captured and analyzed from Wireshark:

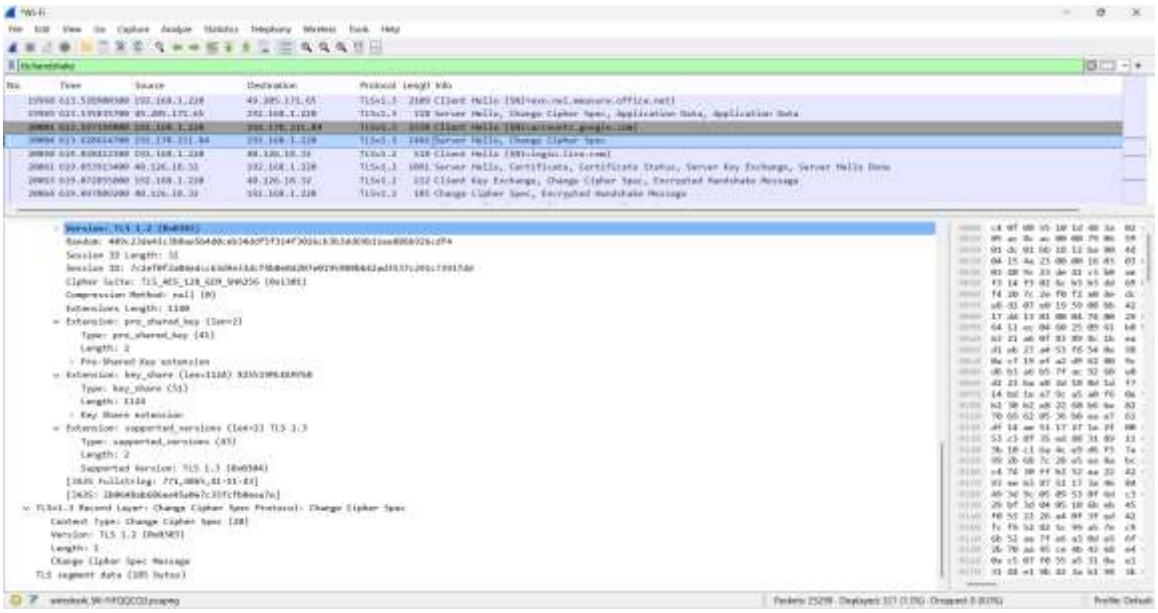
- Certificate Type: X.509 v3
- Signature Algorithm: SHA256 with RSA Encryption
- Issuer: Certificate Authority (CA)
- Subject: bluecloudsoftech.com
- Serial Number: Unique identifier assigned by CA
- Validity: Includes Not Before and Not After dates
- Public Key: RSA public key used for encryption
- Extensions: Subject Alternative Name (SAN), Key Usage, Extended Key Usage
- Certificate Chain: Server certificate + Intermediate CA certificate

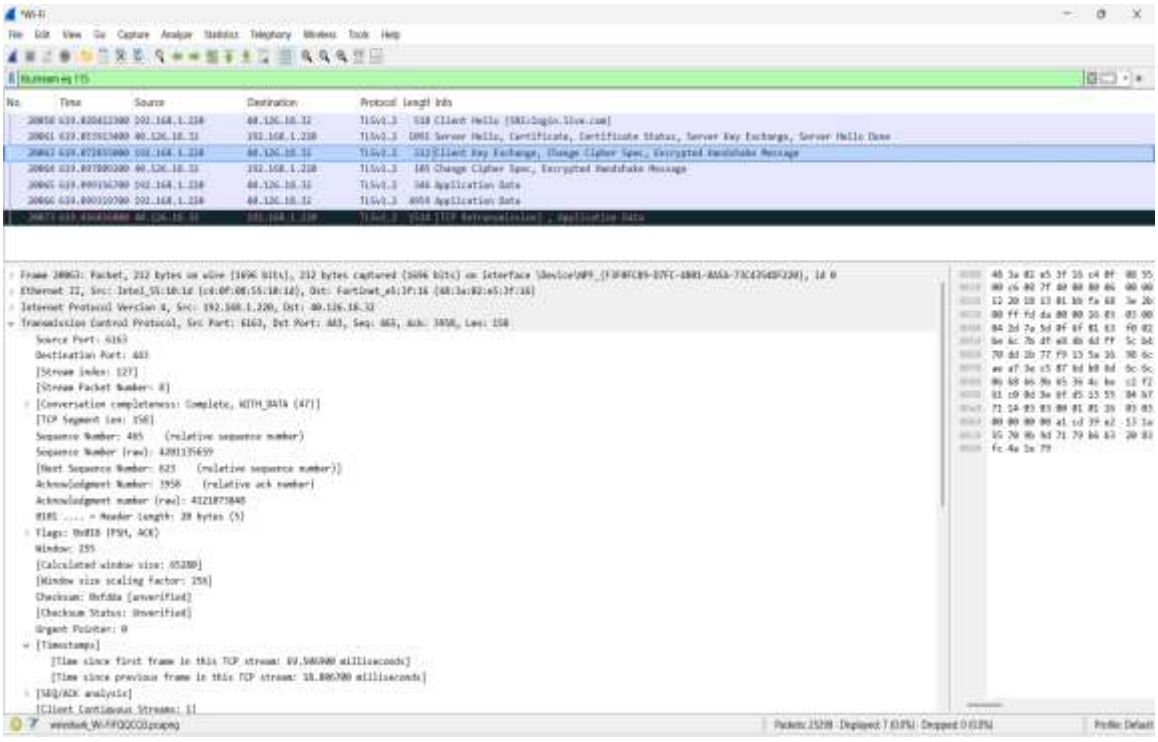
6. Key Findings

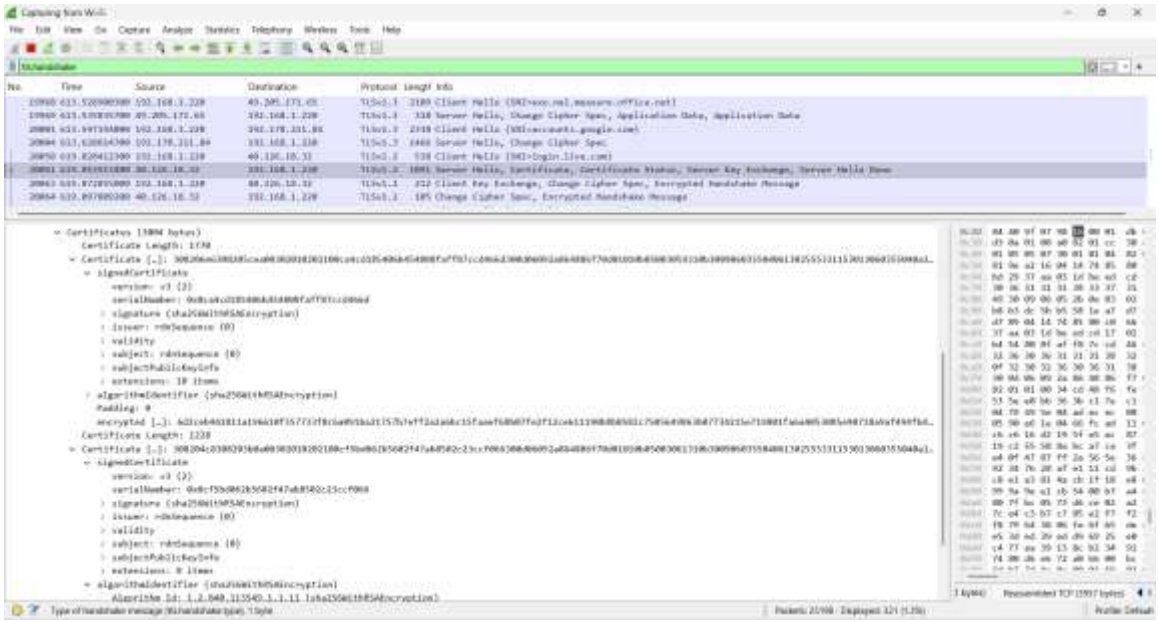
- TLS ensures secure encrypted communication between client and server.
- SSL certificate verifies the identity of the website.
- Packet capture shows handshake metadata but not encrypted data.
- Sensitive information (passwords) is not visible due to encryption.

7. Screenshots









8. Conclusion

TheVSSL/TLS Handshake Analysis of **bluecloudsoftech.com** were successfully completed. The SSL/TLS analysis demonstrated how secure communication is established through certificate verification and encryption. Overall, the activity provided practical understanding of domain infrastructure, DNS security, and secure web communication.